

Alexander Joos

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EDUCATION

Cornell University Ithaca, NY
MEng Computer Science Jan. 2026 – Dec. 2026
Cornell University Ithaca, NY
B.A. Information Science (Data Science) Aug. 2022 – Dec. 2025
National University of Singapore Singapore
Yale NUS College Exchange Program Jan. 2025 – May 2025
Relevant Coursework: Graduate Machine Learning, Graduate Software Engineering, Data Science, Backend Development, Graduate Algorithm Design, Graduate Deep Learning, Statistics, Linear Algebra, Computer Organization

EXPERIENCE

Fidelity Investments June 2026 – Current
Graduate Data Engineering Intern, Database Modernization Merrimack, NH

- Developing low-latency, deterministic distributed brokerage platform & streaming state machine exhaust to SingleStore.
- Building streaming pipelines for capital markets data with Aeron and Kafka, & ETL pipelines from Kafka into SingleStore.
- Developing validation agents & tool-call workflows for agentic legacy-code modernization using TLA+ for design verification.

Fidelity Investments June 2025 – Aug. 2025
Data Engineering Intern, Cloud and Platform Engineering Merrimack, NH

- Built and launched a Flask/AJAX web app on an Ubuntu VM to automate internal database model comparison workflows.
- Supported brokerage platform model analysis across Oracle, PostgreSQL, CockroachDB, & Snowflake databases.
- Implemented Python/SQL backend storage and retrieval logic with JavaScript/HTML/CSS frontend workflows in Flask.
- Deployed the application through Apache, SSL certificate setup, firewall whitelisting, and DEV/PROD rollout.

American Family Insurance: Homesite June 2023 – Aug. 2024
Partner Distribution Intern → Data Analytics Intern Boston, MA

- Built BigQuery pipelines from raw source tables for a Fortune 50 retailer's Partner Management reporting layer in SQL.
- Built 8 Looker dashboards for Partner & Executive teams; automated SQL query generation via Excel/Sheets scripts.
- Conducted Excel, Power Pivot, & Power Query analysis on a top U.S. auto carrier to identify competition feature importance.

PROJECTS

Apache Airflow Anomaly Detection – Agile Open Source Systems & Backend Feature Development May 2026

- Built Airflow-native task-runtime anomaly detection with DAG-file config, custom detectors, & 4 built-in algorithms.
- Implemented backend/task SDK, structured results, task-level observability, CI tests, & example DAGs/docs.

LOB-Driven Optimal Execution Strategy – Quantitative Trading & Market Microstructure May 2026

- Designed TWAP-beating execution strategy using order-book imbalance & spread-threshold signals on tick-level LOB data.
- Backtested on INTC/MSFT/AMZN/GOOG with smoothed grid search and separate parameter fits by spread regime.

LLM Text Watermarking – ML Research Reimplementation May 2026

- Reimplemented Kirchenbauer et al. LLM watermarking in PyTorch with soft green-list logit biasing and z-test detection.

Clustering Analysis on Workplace Gender & Power – NLP & Unsupervised ML Dec. 2024

- Cleaned email bodies into a high-dimensional TF-IDF matrix, capturing sentiment across 15,000+ messages.
- Used DBSCAN, K-means, RiveterNLP, and cross-tabulation approaches to analyze gender, power, and positional clusters.

Old Bailey Verdict Prediction Model – LLM Fine-Tuning & Classification Nov. 2024

- Fine-tuned a Hugging Face LLM on 300+ years of > 100,000 Old Bailey case briefs across 8 legal categories with 71% F1.

SKILLS

Languages: Python, Java, SQL, OCaml, C/C++, R, JavaScript, HTML/CSS

Frameworks & Libraries: Flask, REST APIs, PyTorch, transformers, scikit-learn, pandas, NumPy, matplotlib, SQLAlchemy

Data & Cloud Tools: BigQuery, Vertex AI, Looker, Power BI, SQLite, Oracle, PostgreSQL, Snowflake, DBEaver

Developer Tools: Git, GitHub, Docker Compose, Apache Airflow, Kafka, Bash, Jupyter, Google Colab, Jira, Confluence